

# Cloud Architecture & Infrastructure

Week 5 • Day 2 • Technical Infrastructure & Modeling

TPM Academy



## Day 2 Objectives

- Make the **region + AZ** choice in customer + compliance + cost terms
- Default to **managed services**; defend any self-managed exception
- Take a **multi-tenancy stance** tied to customer evidence
- Choose a **network boundary** with revisit triggers
- Add a **rough order of magnitude (ROM) cost table** — catch 2x surprises before they ship



# Today's Shape

| Clock         | Block                       | Outcome                          |
|---------------|-----------------------------|----------------------------------|
| 09:00 – 09:15 | Opening                     | Pin TMD Section 1 on the wall    |
| 09:15 – 10:30 | Block 1 + Activity 1        | Region + AZ stance               |
| 10:45 – 12:00 | Block 2 + Activity 2        | Managed vs self per component    |
| 13:00 – 14:15 | Block 3 + Activity 3        | Multi-tenancy + network boundary |
| 14:30 – 16:00 | Block 4 + Activity 4 + Wrap | ROM cost + AI sanity check       |

# Block 1 — Region + AZ

Where the customer's data lives matters



# The Cloud Vocabulary Card

| Term                   | What it means                                                                     |
|------------------------|-----------------------------------------------------------------------------------|
| <b>Region</b>          | A geographic cloud location (East US)                                             |
| <b>AZ</b>              | Failure-isolated facility within a region                                         |
| <b>Multi-region</b>    | Across regions; survives whole-region failure                                     |
| <b>Managed service</b> | Cloud provider runs it (Azure Database for PostgreSQL, Blob Storage, Service Bus) |
| <b>Self-managed</b>    | You run it on raw compute                                                         |
| <b>Multi-tenant</b>    | One deployment serves all customers                                               |
| <b>Single-tenant</b>   | Each customer gets their own                                                      |
| <b>Edge / CDN</b>      | Static content cached close to users                                              |

# The Five Decisions

1. **Region(s)** — which?
2. **Multi-AZ** — single, multi-AZ, or multi-region?
3. **Managed vs self-managed** — for each component
4. **Multi-tenancy** — shared / pooled / dedicated / hybrid
5. **Network boundary** — public, VPN, ExpressRoute

Each has a **cost dimension** and a **risk dimension**. Surface both.



## Worked Region + AZ Stance

```
**Region:** East US (primary), West US 2 (warm-DR).  
- Customer base: 92% US; 8% Canada (cross-region OK; +80ms)  
- Compliance: SOC 2 audit retention requires named region  
  
**AZ stance:** Multi-AZ active-active for stateful tiers  
(Postgres primary + 2 read replicas across 3 AZs).  
  
**Trade-off – multi-region active-active vs warm-DR:**  
**Choice:** Warm-DR (Option A).  
**Why:** Active-active doubles cost and adds cross-region  
consistency complexity; multi-AZ within East US  
satisfies our 99.5% availability SLO.  
**Accepted cost:** Full East US outage breaches SLO ~30 min.  
**Revisit trigger:** Second multi-hour East US outage in  
12 months, OR availability SLO tightens to 99.9%+.
```

The warm disaster-recovery (DR) choice is judged against our availability Service Level Objective (SLO).

# Activity 1 — Region + AZ

Quads • 45 minutes

- 5 min: pull customer location facts from Product Requirements Document (PRD) Section 2
- 5 min: pull compliance facts from Technical Concept Document (TCD) Section 3
- 20 min: decide region(s)
- 10 min: decide multi-AZ stance
- 5 min: capture trade-off (Week-4 template)



5 + 5 + 20 + 10 + 5



# Block 2 — Managed vs Self

Default to managed; defend exceptions

# ? The Four-Question Test

Default: **use the managed service**. Reasons to self-manage:

1. Do failure modes match our needs?
2. Is cost reasonable at expected scale?
3. Does compliance allow it?
4. Do we have a specific reason to control this?

**"We want full control"** is a smell. Push: what specific decision do you need that the managed service forecloses?



# The Managed/Self Table

| Component         | Managed option                | Choice       | Why                               |
|-------------------|-------------------------------|--------------|-----------------------------------|
| Postgres          | Azure Database for PostgreSQL | Managed      | Default; team ops capacity finite |
| Event Hubs        | Azure Event Hubs              | Managed      | Same                              |
| Object storage    | Blob Storage                  | Managed      | Industry-standard durability      |
| Custom AI service | None                          | Self-managed | No managed equivalent             |
| Load balancer     | Application Gateway           | Managed      | Trivial to use; standard          |

## Activity 2 — Managed vs Self

Quads • 50 minutes

- 5 min: list each container (TCD Section 2)
- 10 min: identify managed-service option per container
- 25 min: managed vs self per container (4-question test)
- 10 min: capture the table



5 + 10 + 25 + 10

# Block 3 — Tenancy + Network

Stakeholder-touching decisions



# Multi-Tenancy Spectrum

| Stance    | Architecture                       | Pros                | Cons              |
|-----------|------------------------------------|---------------------|-------------------|
| Shared    | One deployment, all tenants        | Cheap               | Noisy neighbor    |
| Pooled    | One deployment + per-tenant limits | Cheap-ish; isolated | Tenant-aware code |
| Dedicated | One deployment per tenant          | Strongest isolation | Expensive         |
| Hybrid    | Most pooled; some dedicated        | Match cost to value | Two patterns      |

**Most B2B SaaS at our stage:** pooled with per-tenant rate limits (revisit Week 4 Day 4).



# Network Boundary

## Public HTTPS

Default for B2B  
SaaS APIs. Standard  
Transport Layer  
Security (TLS);  
nothing else.

## Customer-VPN-only

Some enterprise  
contracts; customer  
comes through their  
VPN tunnel.

## ExpressRoute / Private Link

Large enterprise;  
regulated industries;  
dedicated network  
paths.

**Don't volunteer for VPN-only.** It triples ops complexity. Only choose if a customer requires it.

# Activity 3 — Tenancy + Network

Quads • 50 minutes

- 10 min: customer tenancy expectations (PRD Section 1)
- 10 min: pick stance with revisit trigger
- 25 min: network boundary stance
- 5 min: stakeholder implications (add to TCD Section 6 if missing)



10 + 10 + 25 + 5



# Block 4 — Cost + AI Sanity

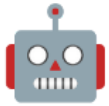
Catch 2x surprises before they ship



# ROM Cost Table

| Component                     | Stance               | Cost / month   | Driver                            |
|-------------------------------|----------------------|----------------|-----------------------------------|
| Azure Database for PostgreSQL | Multi-AZ, 2 vCore GP | 400–700        | Hours; storage; IOPS              |
| Read replica                  | Single-AZ            | 200–350        | Hours                             |
| Event Hubs                    | Standard, 5 TUs      | 700–1100       | Throughput units; storage; egress |
| Blob Storage (audit)          | Standard tier        | \$50/TB-month  | Storage; read/write ops           |
| Application Gateway           | Standard             | \$25 + traffic | Hours + CUs                       |
| Egress bandwidth              |                      | 50–150         | \$/GB out                         |

**Goal: rough order of magnitude.** Engineering will refine. The TPM job: catch the 2x surprise.



# AI Sanity Check

Role: SRE reviewing a feature's cloud topology.

Context: <paste TMD Section 2 sections – region, AZ, managed services, tenancy, network boundary, ROM cost>

Task: Identify the top 3 risks in this topology that could surprise the team within 6 months.

Constraints:

- Be specific to the configurations described
- For each: scenario, likelihood (L/M/H), suggested mitigation
- Do not invent platform features

Format: Numbered – Risk / Scenario / Likelihood / Mitigation.

## Activity 4 — Cost + AI Sanity

Quads • 55 minutes • Wrap

- 25 min: ROM cost table
- 10 min: AI sanity check (3 risks)
- 10 min: adopt / defer / reject
- 10 min: polish Section 2 + provenance note



25 + 10 + 10 + 10 • 15-min wrap



# Day 2 Wrap

## What we built

- Region + AZ stance
- Managed/self table
- Multi-tenancy + network boundary
- ROM cost table
- Technical Modeling Document (TMD) Section 2 drafted

## What opens tomorrow

- **REST & SOAP API fundamentals**
- Resource modeling, idempotency, errors
- When SOAP is still right

**Bring to Day 3:** TMD Sections 1–2 + your data model. APIs ride on top of both tomorrow.

# Questions & Reflection

Where does your topology assume the team has expertise it might not have in 12 months?